

Ammar Vora

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SKILLS

Languages: Python, C/C++, Java, JavaScript, HTML/CSS, Assembly, Verilog HDL

Frameworks & Libraries: PyTorch, OpenCV, NumPy, ROS/ROS2, Node.js, React.js, Express.js, JavaFX

Tools: Git, GitHub, Gitlab, Linux, MongoDB, Google Firestore, Firebase, Google Cloud Platform, Postman, FPGA

EDUCATION

University of Toronto (Toronto, ON, Canada)

September 2021 - April 2026

Bachelor's of Applied Science and Engineering – Computer Engineering with minor in Artificial Intelligence (AI)

Cumulative GPA: 3.81/4.0 (Honors Student, Dean's List)

Relevant Coursework: Programming Fundamentals in C++, Applied Fundamentals of Deep Learning

Activities: President of the University of Toronto Robotics Association, Head Leader for the Orientation Committee

EXPERIENCE

Medical Computer Vision and Robotics Lab (University of Toronto)

May 2023 - August 2023

Undergraduate Researcher

- Developing an image processing pipeline to generate 3D reconstructions of the Larynx to quantify a rare disease
- Fine-tuned hyperparameters while training a CNN based trachea detection model, resulting in **94%** test accuracy
- Conducted data collection and labeled endoscope images using the Computer Vision Annotation Tool (CVAT)

AUToronto, Self-Driving Car Team (GM-SAE Autodrive Challenge II)

September 2022 - August 2023

Member of Lane Detection team

- Collaborated on restructuring and testing a lane detection pipeline, ranking **1st** in the Perception Challenge
- Built a lane line clustering algorithm using OpenCV, leading to a **robust** and **real time** system running at **30 fps**
- Leveraged internal APIs to build a ROS2 node in C++ to report nearby lane lines to the competition scoring system
- Developed rigorous test cases to validate the lane detection pipeline and present results at 4 internal milestones

Autonomous Rover Team (Intelligent Ground Vehicles Competition)

September 2021 - August 2023

Lead of Computer Vision team

- Spearheaded development of the object detection pipeline, resulting in **3rd** place at the design presentation
- Optimized an algorithm to project 2D object detections into 3D in C++, accelerating runtime by **3 times**
- Optimized architecture of the U-Net model in PyTorch, boosting inference speed by **10x** while retaining accuracy
- Designed an automatic pothole dataset generation pipeline in Python, **saving over 50 hours** in data labeling
- Implemented testing-based development and encouraged code reviews, increasing member involvement by **150%**

The University of Toronto Entrepreneurship Hatchery

May 2022 - August 2022

Co-Founder and CTO at FaverIT

- Developed a real-time chat feature in a React Native application and connected it to a NoSQL Firestore database
- Designed and implemented a microservices-based backend architecture using Node.js to ensure scalability
- Integrated JSON Web Tokens alongside a Google OAuth login system to allow users to securely access REST APIs

PROJECTS

Birds - Applied Fundamentals in Deep Learning

May 2023 - August 2023

- Trained a Generative Adversarial Network with CNNs in Pytorch to animate the flying motion of a bird in an image

City Explorer - Software Communication and Design

January 2023 - April 2023

- Designed fast object-oriented APIs and algorithms in C++ that visualize OpenStreetMap data in **5 milliseconds**
- Achieved a path routing time of **less than 10 milliseconds** by implementing Dijkstra's path finding algorithm in C++
- Multi-threaded a Genetic Algorithm for the traveling salesman problem and enhanced runtime speed by **400%**

Rescue Ranger - UofTHacks X (Awarded 1st Place) 🏆

January 2023 - January 2023

- Engineered an autonomous rover and a **full-stack** web application to help with Search and Rescue missions